CareConnect – Changes Since the Original Zip

1. Project structure and setup

- **API reads the root folders:** `sql/`, `mongo/` and `data_generation/` stay at the repo root (brief's suggested layout). Paths fixed in `api/app/services.py`, `api/scripts/bootstrap.py`, `api/scripts/fresh_pings.py` and `api/tests/conftest.py`. *(your edit)*
- **Added `sql/07_api_support.sql`:** `api_sessions`, `api_booking_requests`, `api_refresh_metadata` tables and indexes; checked against the API code. *(your file)*
- **Latest-ping-per-nurse index** `(nurse_id, created_at, _id)` merged into `mongo/01_collections_and_indexes.js`; no separate `04` file. *(your edit)*
- **Docs moved to the root:** `api/docs/*` → `docs/`, duplicates removed, links updated.
- **Docker:** `docker-compose.yml` moved to the root; the API image builds from the repo root (`api/Dockerfile` + `api/Dockerfile.dockerignore`), installs Faker, and the deleted `04` script is no longer run.
- **One root `.env.example`** for every setting (replaces `api/.env.example`).
- **One root `.gitignore`** covering `.env`, `.venv`, `node_modules` and web build/test output; `api/.gitignore` and `web/.gitignore` deleted.

2. Seed data

- **Real names:** Faker generates patient and clinic names; random each run, or repeatable with `CARECONNECT_SEED`.
- **10 of 50 clinics not accepting patients;** clinic 1 always accepts.
- **Audit log:** a mix of credits and debits; timestamps spread over 60 days so the date filter works. Running balances are unchanged.
- **MongoDB seeder:** restored the 40 `MedicalCatalogs` documents and the clearing of old data (lost in your copy); added a `Nurses` collection (names + home area near a clinic) and a physician on each catalog entry.
- **Nurse pings:** every nurse moves around one of the clinics, timestamps spread over the last 100 minutes; every clinic has nurses within 5 km. Shared logic in the new `data_generation/nurse_telemetry.py`.

- **`api/scripts/fresh_pings.py`** is now a live simulator: one new ping per nurse; `--loop N` repeats every N seconds.

3. API (backend)

- **Audit trail sorted newest first.**
- **New `GET /api/v1/catalogs/specialties`** and an exact `specialty` filter on `/catalogs` (for the Reviews specialty chips).
- **Specific error codes instead of generic ones:**

Area	New codes
Booking	`CLINIC_NOT_FOUND`, `PATIENT_ONLY`
Status changes and staff actions	`STAFF_ONLY`, `OTHER_CLINIC`
Records	`OWN_RECORD_ONLY`, `NO_PATIENT_SELECTED`
Dates	`INVALID_DATE_RANGE`, `DATE_OUTSIDE_WINDOW`, `INVALID_MONTH`
Server and databases	`POSTGRES_UNAVAILABLE`, `MONGO_UNAVAILABLE`, `DATABASE_TIMEOUT`, `DATABASE_SETUP_INCOMPLETE`, `DATABASE_ERROR`, `DEMO_MODE_OFF`

4. Frontend

No colours or fonts changed.

Screen	Changes
All screens	A readable message for every API error code above
Browse	No search box on the Patients tab when acting as a patient
Transaction	The form names the one problem instead of a combined message
Audit trail	"Newest first"

Screen	Changes
Analytics	Ranking date picker moved into the Clinic ranking section (top row now 3 cards); date pickers limited to the 30-day window; "Switch to clinic staff…" hint under the disabled Refresh button; clearer message when no clinics are ranked
Map	Nurses listed by name
Reviews	Stars next to the average; "Satisfied patients %"; "Based on N reviews from all clinics"; percentages in the rating distribution and "1 star"; specialty chips replace the search and "Document" boxes; entries show specialty · physician first, friendly field names, ID moved to Additional fields; "Catalog entry X of Y"; same header and top spacing as the other screens on laptop and desktop

5. Tests

- **API tests:** updated for the new error codes; added checks for newest-first audit order and for the specialty list and filter.
- **Browser tests:** added checks for nurse names, the refresh hint, the patient Browse view and the ranking-date position — 18 tests in total, all passing.

6. Documentation

- **Root `README.md` rewritten:** structure, manual and Docker setup, demo walkthrough, assumptions, tech choices, code notes, tests and submission placeholders; Assignment 1 evidence kept.
- **`api/README.md` and `web/README.md`** are now short pointers to the root README; **`docs/inherited_README.md`** deleted.
- **Updated:** `docs/handover_note.md` (new "Assignment 2 integration fixes" section), `docs/mongo_schema_map.json` (`Nurses`, `nurse_name`, `physician`), `docs/api_endpoints.md`, `web/docs/API_CONTRACT.md`, `docs/submission_checklist.md`, `docs/verification.md`.